

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (15 points) Find $f'(x)$ where $f(x) = \frac{2}{x^2} + (1+x^2)\sqrt{x}$.

$$\begin{aligned}
 \text{Solution: } f'(x) &= \frac{d}{dx} \left[\frac{2}{x^2} \right] + \frac{d}{dx} [(1+x^2)\sqrt{x}] \\
 &= -\frac{4}{x^3} + (1+x^2) \frac{d}{dx} [\sqrt{x}] + \sqrt{x} \frac{d}{dx} [1+x^2] \\
 &= -\frac{4}{x^3} + (1+x^2) \left(\frac{1}{2\sqrt{x}} \right) + \sqrt{x} \cdot (2x) \\
 &= -\frac{4}{x^3} + \frac{1}{2\sqrt{x}} + \frac{x^2}{2\sqrt{x}} + 2x\sqrt{x} \\
 &= -\frac{4}{x^3} + \frac{1}{2\sqrt{x}} + \frac{5}{2}x\sqrt{x}
 \end{aligned}$$

Problem 2. (15 points) Compute

$$\begin{aligned}
 &\frac{d}{dx} \left[\frac{x^3 \sin x}{1+x^2} \right] \\
 \text{Solution: } \frac{d}{dx} \left[\frac{x^3 \sin x}{1+x^2} \right] &= \frac{(1+x^2) \frac{d}{dx} [x^3 \sin x] - x^3 \sin x \frac{d}{dx} [1+x^2]}{(1+x^2)^2} \\
 &= \frac{(1+x^2) \left(x^3 \frac{d}{dx} [\sin x] + \sin x \frac{d}{dx} [x^3] \right) - x^3 \sin x \cdot (2x)}{(1+x^2)^2} \\
 &= \frac{(1+x^2) (x^3 \cos x + 3x^2 \sin x) - 2x^4 \sin x}{(1+x^2)^2} \\
 &= \frac{x^2 (x \cos x + 3 \sin x)}{1+x^2} - \frac{2x^4 \sin x}{(1+x^2)^2}
 \end{aligned}$$